

20250728EB_MN (seahorse, RNA, TDP43)

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2025年7月28日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年7月30日 星期三

T2 -> T3

2025年8月1日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年8月4日 星期一

T4 -> T5

2025年8月6日 星期三

T5 -> T6

2025年8月8日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年8月11日 星期一

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add 2mL accutase in 15mL falcon
6. Put the tube into 37C water bath. Flick the cells every 5 min until the accutase went cloudy (~45min this time). (There are many cells attached on the well of 15mL falcon. Will use 50mL next time)
7. pipette the accutase gently and slowly around 20-30 times to break the EBs by checking if there are still visible EBs. (I can still see a lot of undissociated EBs this time, so I take the accutase without visible EBs for the further process.
8. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently.

9. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
10. Filter the cell with 40um filter into a new 50mL tube
11. Spin the cell down at 100rcf, 3.5min. Remove the supernatent.
12. Add 1ml of T7 to resuspend the cells thouroughly to avoid clusters.
13. Add more T7 (~5mL) to resuspend the cells and count it.
14. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info				^
	A	B	C	
1	cell	passage	well#	
2	WT11	P5	6	

Table1			^
	A	B	
1	run1	volume (mL)	
2	accutase	2.5	
3	HI-FBS/CTWM	5	
4	run2	*take some undissolved EB	
5	accutase	1.5	
6	HI-FBS/CTWM	3	

Table3											/
	A	B	C	D	E	F	G	H	I	J	
1		conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume	
2	run1	2.04	94	5	10.2	A	96	20	12	118	
3							96	40	12	235	
4							96	60	12	353	
5						B	24	225	24	2647	
6	run2	0.785	92	5	3.925	C	6	1214	6		

Well2 ^

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												
G												
H												

Well3 ^

	1	2	3	4	5	6
A						
B						
C						
D						

Well4 ^

	1	2	3
A			
B			

2025年8月13日 星期三

T7-> T7

2025年8月14日 星期四

T7 -> BPT8

2025年8月15日 星期五

1/2 BP T8 -> BP T8

2025年8月18日 星期一

1/2 T8 -> 1/2 T8

2025年8月20日 星期三

1/2 T8 -> 1/2 T8

Treat plate B with SPY595 1:5000 at 16:40

add 2.2uL of SPY595 into 230 uL BP T8 -> add 10uL into each well and pipette a bit (except for 1CD)

1CD: add 0.2uL SPY595 into 25uL 015mM glucose NBA (full) -> add 10uL into 1C and 1D

Seahorse for plate A

Well5												
	1	2	3	4	5	6	7	8	9	10	11	12
A	2.5mM glucose 24h	0.15mM glucose 24h	2.5mM glucose 1h incubator	0.15mM glucose 1h incubator	2.5mM glucose preassay	0.15mM glucose preassay						
B												
C												
D												
E												
F												
G												
H												

2025年8月21日 星期四

plate C 24h treatment

Well1			
	1	2	3
A	2.5mM glucose 1h	2.5mM glucose 24h	2.5mM glucose +DMSO 24h
B	0.15mM glucose 1h	0.15mM glucose 24h	2.5mM glucose +2uM MG132 24h

Plate B Imaging

Table2

	A	B	C	D	E	F	G	H
1	condition#		A	B	C	D	E	
2		condition						
3	1	2.5mM glucose 1h						
4	2	0.15mM glucose 1h						
5	3	2.5mM glucose 24h						
6	4	0.15mM glucose 24h						
7	5	2.5mM glucose + MG132	0.07	0.22	0.67	2	6	uM
8	6	0.15mM glucose + MG132	0.07	0.22	0.67	2	6	uM
9	7	2.5mM glucose + tunicamycin	0.04	0.11	0.33	1	3	uM
10	8	0.15mM glucose + tunicamycin	0.04	0.11	0.33	1	3	uM

drug prep (800uL/condition)

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		stock (mM)	working (uM)	dilution	1st dilution	0.07	0.22	0.67	2	6	total volume for dilution 1/condition	total volume for dilution 1	stock volume for dilution 1
2	MG132	10.5	2	5250	100	0.5	1.7	5.1	15.2	45.7	68.2666666667	100	1.0
3						0.04	0.11	0.33	1	3			
4	Tunicamycin	6.12	1	6120	100	0.5	1.4	4.3	13.1	39.2	58.5620915033	100	1.0

condition layout

	1	2	3	4	5	6
A	1	5A	5B	5C	5D	5E
B	2	6A	6B	6C	6D	6E
C	3	7A	7B	7C	7D	7E
D	4	8A	8B	8C	8D	8E

Treatment time											
	A	B	C	D	E	F	G	H	I		
1	#			treat time	JFX650 ONLY	image time (1h)	image time (6h)				
2	1	2.5mM glucose 1h		13:35		14:48	19:27				
3	2	0.15mM glucose 1h		13:37		14:55	19:33				
4	3	2.5mM glucose 24h		8/20 15:55	14:35	15:37					
5	4	0.15mM glucose 24h		8/20 15:55	14:35	15:45					
6	5	2.5mM glucose + MG132	A			15:57					
7			B			16:03					
8			C			16:11					
9			D			16:19					
10			E		10:10-16	16:30					
11	6	0.15mM glucose + MG132	A	11:00-11:06		16:58					
12			B			17:06					
13			C			17:15					
14			D			17:23					
15			E			17:30					
16	7	2.5mM glucose + tunicamycin	A	11:50-11:56		17:50					
17			B			17:56					
18			C			18:05					
19			D			18:12					
20			E			18:20					
21	8	0.15mM glucose + tunicamycin	A	12:40-12:46		18:30					
22			B			18:40					
23			C			18:45					
24			D			18:52					
25			E			18:57					

TDP43 O.C. , 60X, 600x900, 16-bit						^
	A	B	C	D	E	
1	channel	bit	power (%)	exposure time	note	
2	561	16	20	300ms		
3	640	16	10	300ms		

Table4



	A	B	C	D	E	F	G
1	image	treatment	drug	drug_conc entration	time	media	Halo_tim e
2	MN_TDP43Halo_60x_0001	2.5mM glu			1h	full	1h
3	MN_TDP43Halo_60x_0002	0.15mM glu			1h	full	1h
4	MN_TDP43Halo_60x_0003	2.5mM glu			24h	full	1h
5	MN_TDP43Halo_60x_0004	0.15mM glu			24h	full	1h
6	MN_TDP43Halo_60x_0005	2.5mM glu	MG132	0.07	6h	full	6h
7	MN_TDP43Halo_60x_0006	2.5mM glu	MG132	0.22	6h	full	6h
8	MN_TDP43Halo_60x_0007	2.5mM glu	MG132	0.67	6h	full	6h
9	MN_TDP43Halo_60x_0008	2.5mM glu	MG132	2	6h	full	6h
10	MN_TDP43Halo_60x_0009	2.5mM glu	MG132	6	6h	full	6h
11	MN_TDP43Halo_60x_0010	0.15mM glu	MG132	0.07	6h	full	6h
12	MN_TDP43Halo_60x_0011	0.15mM glu	MG132	0.22	6h	full	6h
13	MN_TDP43Halo_60x_0012	0.15mM glu	MG132	0.67	6h	full	6h
14	MN_TDP43Halo_60x_0013	0.15mM glu	MG132	2	6h	full	6h
15	MN_TDP43Halo_60x_0014	0.15mM glu	MG132	6	6h	full	6h
16	MN_TDP43Halo_60x_0015	2.5mM glu	tunicamyci n	0.04	6h	full	6h
17	MN_TDP43Halo_60x_0016	2.5mM glu	tunicamyci n	0.11	6h	full	6h
18	MN_TDP43Halo_60x_0017	2.5mM glu	tunicamyci n	0.33	6h	full	6h
19	MN_TDP43Halo_60x_0018	2.5mM glu	tunicamyci n	1	6h	full	6h
20	MN_TDP43Halo_60x_0019	2.5mM glu	tunicamyci n	3	6h	full	6h
21	MN_TDP43Halo_60x_0020	0.15mM glu	tunicamyci n	0.04	6h	full	6h
22	MN_TDP43Halo_60x_0021	0.15mM glu	tunicamyci n	0.11	6h	full	6h
23	MN_TDP43Halo_60x_0022	0.15mM glu	tunicamyci n	0.33	6h	full	6h
24	MN_TDP43Halo_60x_0023	0.15mM glu	tunicamyci n	1	6h	full	6h
25	MN_TDP43Halo_60x_0024	0.15mM glu	tunicamyci n	3	6h	full	6h
26	MN_TDP43Halo_60x_0025	2.5mM glu			6h	full	6h

27	MN_TDP43Halo_60x_0026	0.15mM glu			6h	full	6h
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